5. AUDITORNE VJEŽBE

~ Geotermalna energija ~

(1)

1. Kruti spremnik volumena 20 m3 sadrži zrak temperature 400 K pod tlakom 1 MPa. Koliko se maksimalnog rada može dobiti iz zraka, kojeg smatramo idealnim plinom (R = 287 J/kgK, cp = 1004 J/kgK), ako potencijalnu i kinetičku energiju zraka zanemarimo? Tlak je okolice 100 kPa, a temperatura 300 K.

$c_v = 717\ J/kgK$

$V = konst. = 20\,m^3$ $\quad R = 287\ J/kgK$ $\quad e_{ks\,ZS} = W_{max}$

$T_1 = 400K$ $\quad c_p = 1004\ J/kgK$ $\quad e_{ks\,ZS} = U - U_{ok} - T_{ok}(S - S_{ok}) + \underbrace{p_{ok}(V - V_{ok})}_{\text{rad potiskivanja okolice}}$

$p_1 = 1\,MPa$ $\quad E_p, E_k \approx 0$

$W_{max} = ?$ $\quad p_{ok} = 100\,kPa$, $T_{ok} = 300K$

$-p_{ok}\int_{V_1}^{V_{ok}} dV$

$$\xi = \frac{w}{e_{ks}} \quad / : q_{dov}$$

$$\underbrace{\frac{\xi}{q_{dov}}}_{q_{dov}} = \eta_t \cdot \frac{1}{e_{ks}} \rightarrow e_{ks} = \eta_{tCKP} \cdot q_{dov}$$

$$e_{ks} = \left(1 - \frac{T_{odv}}{T_{dov}}\right) \cdot q_{dov}$$

T_{ok}, T_1

ZS: $q_{dov} = U$ (unutrašnja kalorična en.) $\rightarrow u = q_{12} = c_v(T_2 - T_1)$ (IZOTERMNO)

W_{max} = maks. rad koji bi se mogao dobiti iz postojećeg stanja sustava bez dovođenja toplinske en. izvana

$$e_{ks} = q_{dov} - T_{odv} \cdot \frac{q_{dov}}{T_{dov}}; \qquad ds = \frac{dq}{T}$$

promjena entropije

Maks. rad se dobije izjednačavanjem stanja sustava sa stanjem okolice povratljivim procesom

stanje minimalne energije koju sustav može postići (zato oduzimamo U_{ok} i T_{ok})

Masa zraka u spremniku: $pV = mRT$

$$m = \frac{p_1 V_1}{R T_1} \rightarrow m = 174.2\ kg \qquad V = \frac{R T_{ok}}{p_{ok}}$$

odnosno: $e_{ks\,ZS} = U - U_{ok} - T_{ok}(S - S_{ok}) + p_{ok}(V - V_{ok})$ $\quad R\frac{T_1}{p_1}$

$$\dot{w}_{max} = c_v \cdot (T_1 - T_{ok}) - T_{ok}\left(c_p \cdot \ln\frac{T_1}{T_{ok}} - R\ln\frac{p_1}{p_{ok}}\right) + p_{ok}\left(\cancel{V_1} - \frac{R T_{ok}}{p_{ok}}\right)$$

$$\dot{W}_{max} = \dots = \cancel{2.097\ MJ/kg} = 108.68\ kJ/kg$$

$$W_{MAX} = m \cdot \dot{w} \Rightarrow \boxed{W_{MAX} = 18.93\ MJ}$$

ZADATAK 2.)

2. Zrak temperature 400 K pod tlakom 1 MPa ekspandira u plinskoj turbini na tlak i temperaturu okolice (100 kPa, 300 K). Koliko se maksimalnog rada može dobiti iz zraka, kojeg smatramo idealnim plinom (R = 287 J/kgK, cp = 1004 J/kgK) ako potencijalnu i kinetičku energiju zraka zanemarimo?

$T_1 = 400K$ $\quad R = 287\ J/kgK$

$p_1 = 10^6\ Pa$ $\quad c_p = 1004\ J/kgK$

c_v $\quad c_v = 717\ J/kgK$

$p_{ok} = 10^5\ Pa$ $\quad E_k, E_p \approx \emptyset$

$T_{ok} = 300K$

$W_{max} = ?$ $\quad pV = mRT$

$M = \frac{p_1 V_1}{RT_1} \rightarrow \cancel{m =}$ $\quad V = ?$

$e_{kos\,os} = W_{max} = \underbrace{h_1 - h_{ok}}_{\text{entalpija}} - T_{ok}(s - s_{ok})$

$c_p(T_2 - T_1)$

$e_{kos\,os} = c_p(T_1 - T_{ok}) - T_{ok}(s - s_{ok})$

$= c_p(T_1 - T_{ok}) - T_{ok}\left(c_p \ln\frac{400}{300} - R\ln\frac{10^6}{10^5}\right)$

$e_{kos\,os} = 212\ kJ/kg$

$\downarrow$

$\boxed{W_{max} = 212\ kJ/kg}$

Zad. 3.)

3. Idealni plin u zatvorenom spremniku ekspandira iz stanja 1 (500 kPa, 500 K) u stanje 2 (200 kPa, 400 K). Tlak je okolice 100 kPa, a temperatura 300 K. Izračunati maksimalni rad koji plin obavi prelaskom iz stanja 1 u stanje 2. cv = 717 J/kgK, R = 287 J/kgK.

$pV = RT$

$V = konst.$ $\quad p_{ok} = 10^5\ Pa$

$p_1 = 500\ kPa$ $\quad T_{ok} = 300K$

$T_1 = 500K$ $\quad W_{max}\ (1-2) = ?$

$p_2 = 200\ kPa$ $\quad c_v = 717\ J/kgK$

$T_2 = 400K$ $\quad R = 287\ J/kgK$

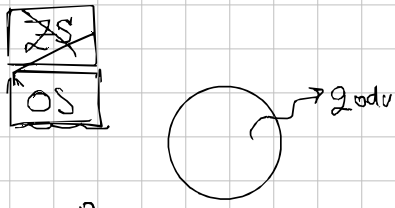
$w_{max} = c_v(T_1 - T_2) - T_{ok}(s - s_{ok}) + R\,p_{ok}\left(\frac{T_1}{p_1} - \frac{T_2}{p_2}\right)$

$w_{max} = c_v(T_1 - T_2) - T_{ok}\left(c_p \cdot \ln\frac{T_1}{T_2} - R\ln\frac{p_1}{p_2}\right) + R\,p_{ok}\left(\frac{T_1}{p_1} - \frac{T_2}{p_2}\right)$

$\boxed{w_{max} = 54681.8\ J/kg}$

Zadatak 4.)

4. Zrak ekspandira u plinskoj turbini od tlaka 7 bara i temperature 600 °C na tlak 1 bar i 250 °C. Za vrijeme ekspanzije, u okolicu, tlaka 1 bar i temperature 15 °C, prelazi 9 kJ/kg toplinske energije. Zanemarujući promjene kinetičke i potencijalne energije, odredite gubitak mehaničkog rada izazvan realnošću procesa u turbini. Pretpostaviti da je zrak idealni plin (cp = 1004 J/kgK, R = 287 J/kgK).



$p_1 = 7 \times 10^5$ Pa $\quad p_{ok} = 10^5$ Pa

$T_1 = 873.15$ K $\quad T_{ok} = 288.15$ K

$p_2 = 10^5$ Pa $\quad q_{odv} = 9$ kJ/kg $= -9000$ $\quad E_k, E_p = \emptyset$

$c_p = 1004$ J/kgK

$T_2 = 523.15$ K $\quad W_{gub\ mehr} = ?$

$R = 287$ J/kgK

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1. način

$$W_{mehr\ gub} = T_{ok} \cdot \Delta S_{uk} \Rightarrow \Delta S_{uk} = \Delta S_s + \Delta S_{ok}$$

$$\Delta S_s = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

$$\Delta S_{ok} = \frac{q}{T_{ok}} = \frac{q_{odv}}{T_{ok}} = -\frac{q_{odv}}{T_{ok}}$$

$$c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} - \frac{q_{odv}}{T_{ok}} \quad (q_{odv} = -9000)$$

$$\Delta S_{uk} = 75.42 \text{ J/kgK} \xrightarrow{\cdot T_{ok}} \boxed{W_{mehr\ gub} = 21.72 \text{ kJ/kg}}$$

2. način

Gubitak mehr. rad $(T_{ok} \Delta S_{uk})$ = W_{max} − W_{real}

eks $\quad$ I. st. termodinam.

(OS)

$$W_{max} = h - h_{ok} - T_{ok}(s - s_{ok}) = c_p(T_1 - T_2) - T_{ok}(s_1 - s_2)$$

$$W_{max} = c_p(T_1 - T_2) - T_{ok}\left(c_p \cdot \ln \frac{T_1}{T_2} - R \ln \frac{p_1}{p_2}\right) = \dots \Rightarrow \boxed{W_{max} = 364.1 \text{ kJ/kg}}$$

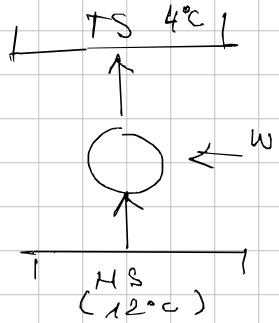
$$q_{12} + h_1 = w_{12} + h_2$$

$$w_{12} = \underbrace{q_{12}}_{q_{dov}} + h_1 - h_2 = \underbrace{q_{dov}}_{-q_{odv}} + \underbrace{c_p(T_1 - T_2)}_{\Delta h = w_{t12}\ \text{adiabatski}} \Rightarrow \boxed{w_{12} = 342.4 \text{ kJ/kg}}$$

$$W_{gub} = 364.1 - 342.4 \longrightarrow \boxed{W_{gub} = 21.7 \text{ kJ/kg}}$$

Zadatak 5.)

Toplinska pumpa koristi podzemnu vodu na 12 °C kao izvor energije. Izlazna temperatura podzemne vode je 4 °C. Proces je izveden s Freonom 12 između tlakova 0,1 MPa i 1,0 MPa. Koliki minimalni protok podzemne vode je nužan da bi toplinska pumpa dobavljala 60 MJ/h? Koliko iznosi potrebna snaga kompresora? Poznate su još vrijednosti entalpija za freon 12 (prema slici): ulaz u kompresor h1 = 174,15 kJ/kg, ulaz u evaporator (isparivač) h3 = h4 = 76,26 kJ/kg, ulaz u kondenzator h2 = 215 kJ/kg. Za specifični toplinski kapacitet vode uzeti cp = 4186 J/kg·K.



Freon 12

Otvoreni sustav

$T_{dov} = 285\,K$

$T_{odv} = 277\,K$

$\dot{Q} = 60\ MJ/h$

$p_{f_1} = 0.1\,MPa = p_1$

$P_k = ?$

$p_{f_2} = 1\,MPa = p_2$

$h_1 = 174.15\,kJ/kg$

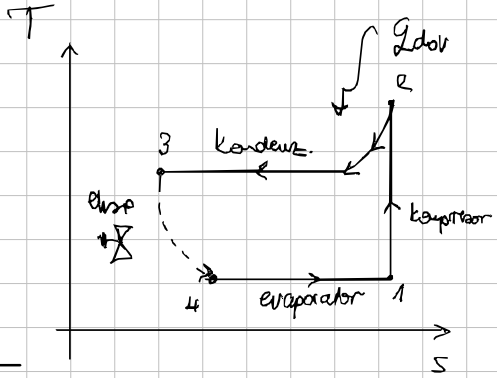
protok$_{min}$ = ?

$h_3 = h_4 = 76.26\,kJ/kg$

$\dot{m}_v = ?$

$h_2 = 215\,kJ/kg$

$c_p = 4186\ J/kgK$



Kondenzator: $q_{23} + h_2 = w_{23} + h_3$ ali $w_{23} = \emptyset$

$Q_{23} = \dot{m}_F \cdot (h_3 - h_2) \longrightarrow \dot{m}_F = \frac{\dot{Q}_{23}}{h_3 - h_2} \Rightarrow \dot{m}_F = 0.12\ kg/s$

$Q_{23} = Q_{dov} = -60\,MJ/h$

$\cancel{q_{12}}^{\emptyset} \cdot h_1 - h_2 = w_{12} \longrightarrow w_k = -40.85\,kJ/kg \longrightarrow P_k = \dot{m}\, w_k$

$\boxed{P_k = -4.9\ W}$

1. GST $\quad Q_{kv} + \sum m_{ulaz} \cdot h_{ulaz} = \sum m_{izlaz} \cdot h_{izlaz} + W_{kv}$

$\dot{m}_F \cdot h_4 + \dot{m}_v \cdot h_{ul} = \dot{m}_F \cdot h_1 + \dot{m}_v \cdot h_{iz}$

$\dot{m}_v = \dot{m}_F \frac{h_1 - h_4}{h_{iz} - h_{ul}} = \dot{m}_F \frac{h_1 - h_4}{-c_p (T_{iz} - T_{ul})} \Longrightarrow \boxed{\dot{m}_v = 0.351\ kg/s}$

Zadatak 6.)

Binarna geotermalna TE u Costa Rici snage 15 MWe koristi Rankineov kružni proces u kome je specifična entalpija na ulazu u turbinu 530 kJ/kg, na izlazu iz turbine 453 kJ/kg, na izlazu iz kondenzatora 15,2 kJ/kg i na izlazu iz pojne pumpe 18,2 kJ/kg. Gubici električnog generatora su zanemarivi.

a) Koliki je termodinamički stupanj djelovanja?

b) Koliki treba biti maseni protok radnog medija u kružnom procesu?

$P = 15\,MW_e$

Rankin

$h_2 = 530\ kJ/kg$

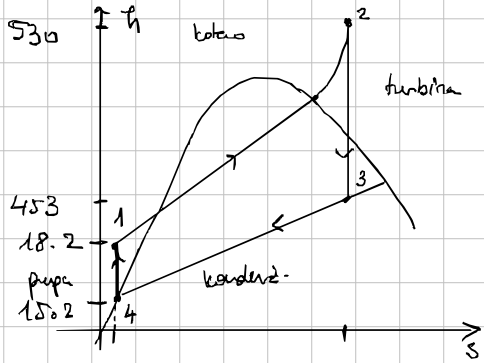
$h_3 = 453\ kJ/kg$

$h_4 = 15.2\ kJ/kg$

$h_1 = 18.2\ kJ/kg$

$\eta_t = ?$

$\dot{m}_F = ?$



$q_{12} = h_2 - h_1 \quad \Rightarrow q_{dov}$

$w_t = h_2 - h_3$

$w_p = h_1 - h_4$

$$\eta_t = \frac{w_t - |w_p|}{q_{dov}}$$

$$\eta_t = \frac{h_2 - h_3 - |h_1 - h_4|}{h_2 - h_1} \longrightarrow \boxed{\eta_t = 0.145}$$

$$\dot{m}_F = \frac{P}{\dot{w}} = \cancel{\frac{P}{w_t - |w_p|}} = \frac{P}{w_t} = \boxed{194.8\ kg/s}$$

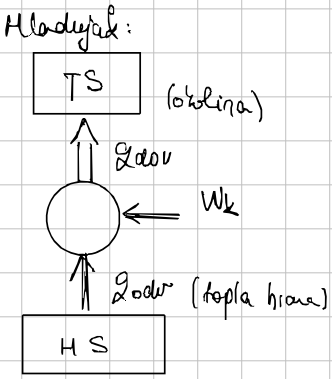
ZADACI PPT

Zadatak 1.) HLADNJAK

Hladnjak koristi rashladno sredstvo-134a kao radni medij u kružnom procesu s idealnom kompresijom među tlakovima 0.14 i 0.8 MPa. Ako je maseni protok radnog medija 0.05 kg/s, odredite: a) brzinu odvođenja topline iz hlađenog prostora i snagu kompresora; b) brzinu odbacivanja topline u okolicu; c) faktor preobrazbe hladnjaka. Podaci za radni medij dani su u nastavku (sljedeći slajd).

- za tlak p1 = 0.14 MPa: h1 = h" = 239.19 kJ/kg

  s1 = s" = 0.94467 kJ/kg·K
- za tlak p2 = 0.8 MPa: s2 = s1 --> h2 = 275.40 kJ/kg
- za tlak p3 = 0.8 MPa: h3 = h' = 95.48 kJ/kg
- h4 =~ h3 (prigušenje) --> h4 = 95.48 kJ/kg



$p_1 = 0.14$ MPa

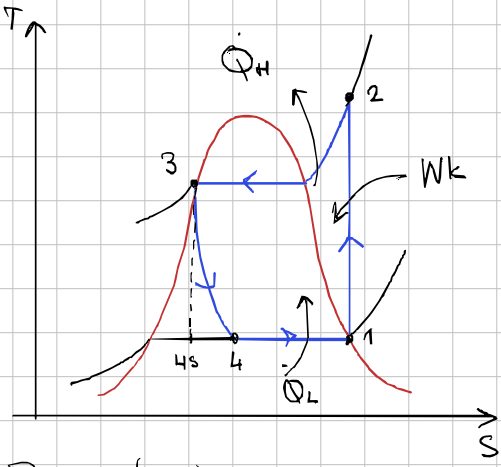
$p_2 = 0.8$ MPa

$\dot{m}_m = 0{,}05$ kg/s

a) $\dot{Q}$ [J/s] = ?

P_k = ? [W = J/s]

„topla" toplina je Q_{odv} jer ju dovodimo u sustav (iz tople hrane)



$\dot{Q}_{odv}$ => 4-1

$q_{41} + h_4 + \emptyset + \emptyset = \emptyset + h_1$

$q_{41} = h_1 - h_4 \quad / \cdot \dot{m}_m$

$\dot{Q}_{odv} = \dot{m}_m (h_1 - h_4)$

$\boxed{\dot{Q}_{odv} = 7.19 \text{ kJ/s}}$

P_k je (1-2)

$\emptyset + h_1 + \emptyset + \emptyset = W_k + h_2 + \emptyset + \emptyset \longrightarrow W_k = h_1 - h_2 = P_k = \dot{m}_m (h_1 - h_2)$

$\boxed{P_k = -1.81 \text{ kW}}$

b) Brzina odbacivanja topline u okolicu => $\dot{Q}_{dov}$ => (2-3)

$q_{dov} + h_2 = h_3 \longrightarrow q_{dov} = h_3 - h_2 \longrightarrow \dot{Q}_{dov} = \dot{m}_m (h_3 - h_2)$

$\boxed{\dot{Q}_{dov} = -8.996 \text{ kJ/s}}$

c) faktor preobrazbe hladnjaka

$COP_{HL} = \left| \frac{Q_{odv}}{W_k} \right| = \left| \frac{\dot{Q}_{odv}}{P_k} \right| \Rightarrow \boxed{COP_{HL} = 3.97}$ > 1 jer je LIKP

Zadatak 2.) HLADNJAK

Rashladno sredstvo-134a (radni medij) ulazi u kompresor hladnjaka kao pregrijana para na tlaku 0.14 MPa i temperaturi -10°C i napušta kompresor na tlaku 0.8 MPa i temperaturi 50°C. Rashladno sredstvo u kondenzatoru se hladi do temperature 26°C i tlaka 0.72 MPa. Maseni protok rashladnog sredstva je 0.05 kg/s. Prigušenje se odvija do tlaka 0.15 MPa.

Odredite: a) brzinu odvođenja topline iz hlađenog prostora i snagu kompresora; b) izentropsku (unutrašnju) efikasnost kompresora;

c) faktor preobrazbe. 10 Podaci za radni medij:

- za tlak p1 = 0.14 Mpa i temp. T1 = – 10°C → h1 = 246.37 kJ/kg
- za tlak p2 = 0.8 Mpa i temp. T2 = 50°C → h2 = 286.71 kJ/kg
- za tlak p3 = 0.72 Mpa i temp. T3 = 26°C → h3 ≈ h' = 87.83 kJ/kg
- h4 ≈ h3 (prigušenje) → h4 = 87.83 kJ/kg
- p2s = 0.8 MPa, s2s = s1 = 0.9724 kJ/kgK, h2s = 284.20 kJ/kg

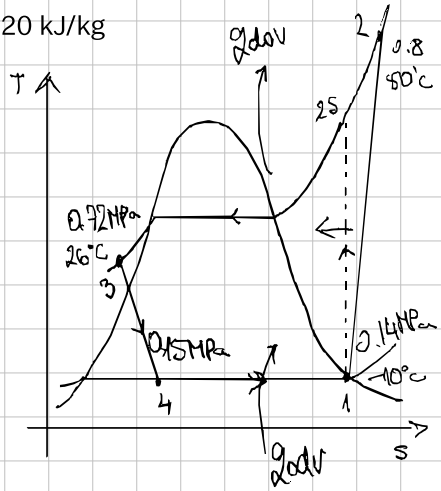
$p_1 = 0.14\,MPa$, $\vartheta_1 = -10°C \Rightarrow T_1 = 263.15\,K$

$p_2 = 0.8\,MPa$, $\vartheta_2 = 50°C \Rightarrow T_2 = 323.15\,K$

$p_3 = 0.72\,MPa$, $\vartheta_3 = 26°C \Rightarrow T_3 = 299.15\,K$ (3)

$\dot{m}_m = 0{,}05\,kg/s$

$p_{gušenja} = 0.15\,MPa$ (p_4)



a) $\dot{Q}_{odv} = ?$ $P_k = ?$

$q_{41} = h_1 - h_4 \longrightarrow \boxed{\dot{Q}_{41} = \dot{m}_m (h_1 - h_4) = 7.93\,kJ/s}$

$w_k = h_1 - h_2 \longrightarrow \boxed{P_k = \dot{m}_m (h_1 - h_2) = -2.02\,kW}$

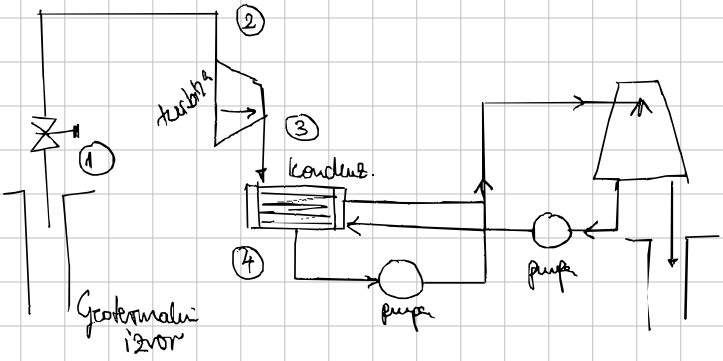
b) $\boxed{\eta_{izk} = \left|\frac{W_{ideal(k)}}{W_{real(k)}}\right| = \left|\frac{h_1 - h_{2s}}{h_1 - h_2}\right| = 0.938}$ kao pumpa

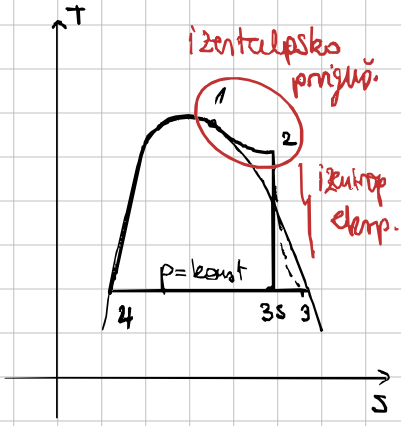
c) $COP_{HL} = \left|\frac{q_{odv}}{w_k}\right| = \left|\frac{\dot{Q}_{41}}{P_k}\right| \rightarrow \boxed{COP_{HL} = 3.93}$ ✓

Zadatak 3.) ZASIĆENA PARA

Geotermalni izvor daje zasićenu paru protoka 25 kg/s na tlaku 2 MPa (212.38 °C, 2798.3 kJ/kg, 6.339 kJ/kgK). Nakon izentalpskog prigušenja na ulaznom ventilu tlak pare je reduciran na 700 kPa (179.54 °C, 6.7869 kJ/kgK). Izentropska efikasnost turbine je 80%, a tlak u kondenzatoru je 20 kPa (60 °C, 251.42 kJ/kg, 0.832 kJ/kgK, 2608.9 kJ/kg, 7.9072 kJ/kgK).

Odrediti sadržaj pare nakon ekspanzije, snagu elektrane, termodinamički i eksergetski stupanj djelovanja. Rad pumpanja zanemariti. Parametri okoline su 20 °C i 101,325 kPa (84 kJ/kg, 0.2965 kJ/kgK). Nacrtati i označiti funkcionalnu shemu i T-s dijagram procesa.





$p_1 = 2 \times 10^6$ Pa

$T_1 = 485.53$ K

$h_1 = 2798.3$ kJ/kg

$s_1 = 6.339$ kJ/kgK

$p_2 = 7 \times 10^5$ Pa

$T_2 = 452.69$

$\boxed{h_2 = h_1}$

$s_2 = 6.7869$ kJ/kgK

$\eta_{iz} = 0.8 \rightarrow \frac{\text{ideal}}{\text{real}}$

$p_3 = 2 \times 10^4$ Pa $= p_4$

$T_3 = 333.15$ K

$h' = 251.42$ kJ/kg

$h'' = 2608.9$ kJ/kg

$s' = 0.832$ kJ/kg

$s'' = 7.9072$ kJ/kgK

$T_{ok} = 293.15$ K

$p_{ok} = 101.325$ kPa

$h_{ok} = 84$ kJ/kg

$s_{ok} = 0.2965$ kJ/kgK

$x_3 = ?$ $P_{el} = ?$ $\eta_t = ?$ $\zeta = ?$

a) $x_3 = ?$ sadržaj pare:

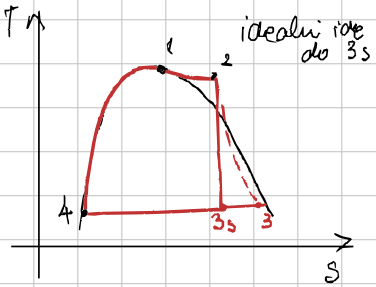
$$x = \frac{s_{3s} - s'}{s'' - s'} \longrightarrow x = \frac{6.7869 - 0.832}{7.9072 - 0.832} \Rightarrow x_3 = 0.8417$$

idealna ekspanzija

$P_{el} = ?$ → dobije se iz snage turbine i efikasnosti generatora

pretpostavljamo $\eta_g = 1$

tražimo realni rad turbine!



$q_{12} = \emptyset$

$w_t = h_2 - h_3$ → nemamo h_3

$$\eta_{it} = \frac{w_{real\,t}}{w_{ideal\,t}} = \frac{h_2 - h_3}{h_2 - h_{3s}}$$

budući da se u zad. radi o realnom procesu u turbini, izentropski korak je samo pomoć!

→ $\eta_{it} = 0.8$ (zadano) ---→ prvo trebam h_{3s}

$h_{3s} = h' + x(h'' - h') \rightarrow h_{3s} = 2235.47 \frac{kJ}{kg}$

$\Rightarrow h_3 = 2348.14$ kJ/kg

$\longrightarrow P_{el} = \dot{m} \cdot w_t \longrightarrow \boxed{P_{el} = 11.267 \text{ MW}}$

$$x_3\,(real) = \frac{h_3 - h'}{h'' - h'} \longrightarrow \boxed{x_3 = 0.889}$$

$\eta_t = ?$

$$\eta_t = \frac{w_t}{q_{dov}} = \frac{h_2 - h_3}{h_1 - h_{ok}} \qquad \boxed{\eta_t = 0.1659}$$

WHY!!!

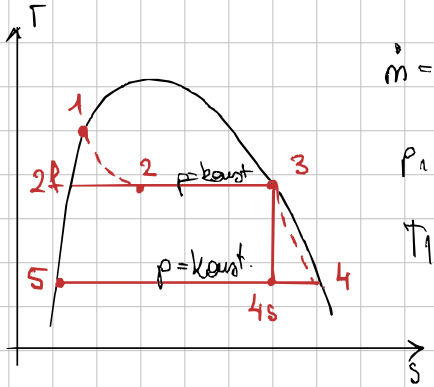
$\xi = ?$

$$\frac{w}{e_{kss}} \rightarrow \frac{w_t}{e_{kss\,os}} = \frac{h_2 - h_3}{h_2 - h' - T_{ok}(s_2 - s')} \Rightarrow \boxed{\xi = 0.562}$$

Zadatak 4.) FLASHING

Geotermalni izvor daje vodu protoka 25 kg/s na tlaku 4 MPa i temperaturi 200 °C (853.27 kJ/kg, 2.3267 kJ/kgK). Nakon izentalpskog prigušenja na ulaznom ventilu tlak je reduciran na 100 kPa (417.50 kJ/kg, 1.3028 kJ/kgK, 2674.9 kJ/kg, 7.3588 kJ/kgK). Izentropska efikasnost turbine je 85%, a tlak u kondenzatoru je 20 kPa (60 °C, 251.42 kJ/kg, 0.832 kJ/kgK, 2608.9 kJ/kg, 7.9072 kJ/kgK).

Odrediti sadržaj pare u separatoru, snagu elektrane i termodinamički stupanj djelovanja postrojenja. Parametri okoline su 20 °C i 101,325 kPa (84 kJ/kg, 0.2965 kJ/kgK). Nacrtati i označiti funkcionalnu shemu i T-s dijagram procesa.



$\dot{m} = 25\,kg/s$

$p_1 = 4\times10^6\,Pa$

$T_1 = 473.15\,K$

$h_1 = 853.27\,kJ/kg$

$s_1 = 2.3267\,kJ/kgK$

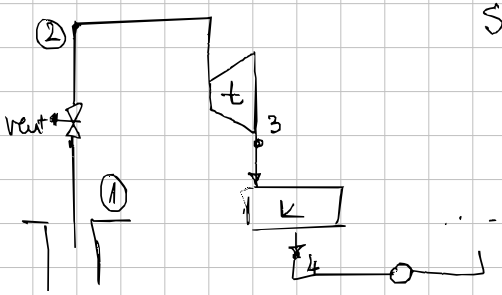
$p_2 = 10^5\,Pa = p_3$

$h' = 417.5\,kJ/kg$

$s' = 1.3028\,kJ/kgK$

$h'' = 2674.9\,kJ/kg$

$s'' = 7.3588\,kJ/kgK$



$\eta_{iT} = 0.85$

$p_4 = p_5 = 2\times10^4\,Pa$

$T_4 = 333.15\,K$

$h' = 251.42\,kJ/kg$

$s' = 0.832\,kJ/kgK$

$h'' = 2608.9\,kJ/kg$

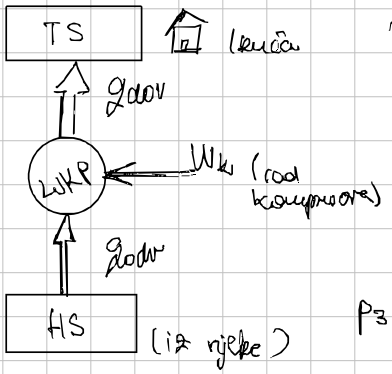
$s'' = 7.9072\,kJ/kgK$

$T_{okl} = 293.15\,K$ $\quad p_{okl} = 101.325\,kPa$

$h_{okl} = 84\,kJ/kg$ $\quad s_{okl} = 0.2965\,kJ/kgK$

Zadatak 5.) TOPLINSKA PUMPA

Toplinska pumpa koristi podzemnu vodu na 12 °C kao izvor energije. Izlazna temperatura podzemne vode je 4 °C. Proces je izveden s freonom 12 između tlakova 0.1 MPa i 1.0 MPa. Koliki minimalni protok podzemne vode je nužan da bi toplinska pumpa dobavljala 60 MJ/h? Koliko iznosi potrebna snaga kompresora? Poznate su još vrijednosti entalpija za freon 12: ulaz u kompresor h1 = 174,15 kJ/kg, ulaz u evaporator (isparivač) h3 = h4 = 76.26 kJ/kg, ulaz u kondenzator h2 = 215 kJ/kg. Za specifični toplinski kapacitet vode uzeti cp = 4186 J/kgK.



$t_{ulz} = 12°C \rightarrow T_{ul} = 285.15K$

$t_{iz} = 4°C \qquad T_{iz}\ 277.15K$

$h_1 = 174.15\ kJ/kg$

$h_3 = h_4 = 76.26\ kJ/kg$

$h_2 = 215\ kJ/kg$

$c_p = 4186\ J/kgK$

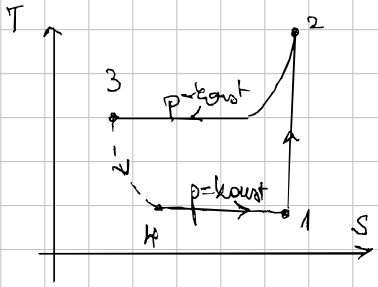
$p_3 = p_2 = 1\ MPa$

$p_4 = p_1 = 0.1\ MPa$

$\dot{m} = ?$

$\dot{Q} = \frac{60\ MJ}{h} = 16.67\ kW$

$P_k = ?$



$\dot{Q} = q_{odv} \cdot \dot{m}$

$q_{odv} = q_{23} \rightarrow q_{23} = h_3 - h_2 \rightarrow q_{odv} = -138.74\ kJ/kg$

$\Rightarrow \dot{m} = \frac{\dot{Q}}{q_{odv}} \Rightarrow \dot{m}_F = 0.12\ kg/s$ (maseni protok freona)

$P_k = \left[\frac{J}{s}\right] = W_k \cdot \dot{m}$; $w_k = h_1 - h_2 \rightarrow w_k = -40.85\ kJ/kg$

$\rightarrow \boxed{P_k = -4.9\ kW}$

ISPARIVAČ je izmjenjivač topline za koji uzimamo da sva toplina oduzeta primarnoj strani (podzemnoj vodi) prelazi na sekund. str. (radni medij ⇒ FREON ; koji grijemo)

$\dot{m}_v (h_{ul} - h_{iz}) = \dot{m}_f (h_1 - h_4)$

raspisivanje 1. GST za OS, tj. kontrolni volumen;

primjena na isparivač $4 \rightarrow 1$: $Q_{KV} + \underbrace{\sum \dot{m}_u h_u}_{\text{ulazne}} = \underbrace{\sum \dot{m}_i \cdot h_i}_{\text{izlazne}} + W_{KV}$ veličine

$\Rightarrow \dot{m}_F \cdot h_4 + \dot{m}_V \cdot h_{ul} = \dot{m}_F \cdot h_1 + \dot{m}_V h_{iz}$

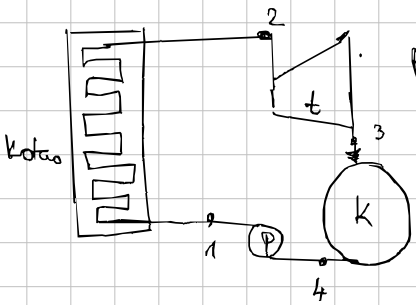
odred: $\dot{m}_F (h_1 - h_4) = \dot{m}_V (h_{ul} - h_{iz}) \longrightarrow \dot{m}_V = \dot{m}_F \underbrace{\frac{h_1 - h_4}{h_{ul} - h_{iz}}}$

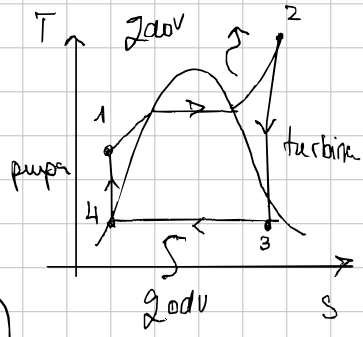
$h_{ul} - h_{iz} = -c_p (T_{iz} - T_{ul}) = c_p (T_{ul} - T_{iz})$ $\Rightarrow$ $\boxed{\dot{m}_V = 0.35 \text{ kg/s}}$

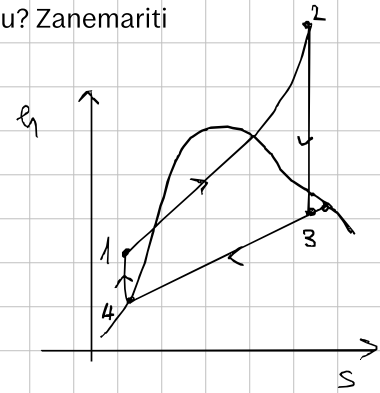
Zadatak 6.) BINARNA → Rankin

Binarna geotermalna TE u Costa Rici snage 15 MWe koristi Rankineov kružni proces u kome je specifična entalpija na ulazu u turbinu 530 kJ/kg, na izlazu iz turbine 453 kJ/kg, na izlazu iz kondenzatora 15.2kJ/kg i na izlazu iz pojne pumpe 18.2 kJ/kg. a) Koliki je termodinamički stupanj djelovanja? b) Koliki treba biti maseni protok radnog medija u kružnom procesu? Zanemariti unutrašnji stupanj djelovanja generatora

$P = 15\,MWe$







$h_2 = 530\ kJ/kg$

$h_3 = 453\ kJ/kg$

$h_4 = 15.2\ kJ/kg$

$h_1 = 18.2\ kJ/kg$

$w_t = h_2 - h_3$

$w_p = h_4 - h_1$

$q_{dov} = h_2 - h_1$

a) $\eta_t = ?$

b) $\dot{m} = ?$

$\eta_g = 1$

$$\eta_t = \frac{w_t - |w_p|}{q_{dov}}$$

$$\dot{m} = \frac{P}{w_t} = \frac{15 \times 10^6}{h_2 - h_3}$$

$$\boxed{\eta_t = 0.1446}$$

$$\boxed{\dot{m} = 194.81\ kg/s}$$